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Overview

This report documents the technical implementation of the Database Schema Refactoring and User Interface (UI) development for the **Classroom Schedule View**. The primary objective was to establish a robust relationship between Classrooms and Enrollments to ensure accurate data retrieval and a clear display of classroom information.

Changes Implemented

1. Database Schema & Migration

- **Entity Relationship Mapping:** Implemented a **Bidirectional Mapping** between `Classroom` and `Enrollment` using `@OneToMany` in the Classroom entity and `@ManyToOne` in the Enrollment entity.
- **Flyway Migration:** Utilized **Flyway** for database version control, ensuring the schema is up-to-date and validated.
- **Data Integrity:** Applied `@Column(nullable = false)` and `@UniqueConstraint` to ensure data safety and prevent duplicate enrollment records for the same student and course.
- **Connection Pooling:** Integrated **HikariCP** to manage database connections between Java and MySQL efficiently with high stability.

2. Backend Logic & Model Refactoring

- **Model Enhancements:**
 - Updated `Enrollment.java` by adding the `classroom` field along with its respective Getters and Setters to support data mapping.
 - Updated `Classroom.java` to include a `List<Enrollment>` to enable fetching all students assigned to a specific room.
- **Controller Implementation:** Transitioned from `@RestController` to a standard `@Controller` to render HTML templates via the Thymeleaf Engine and pass data through the `Model` object.
- **Repository:** Created and configured `ClassroomRepository` to manage classroom data and support course-related queries.

3. UI/UX Development

- **Bootstrap Integration:** Incorporated **Bootstrap 5.3.0** to create a structured, professional, and responsive table layout.
- **Classroom View Interface:** Developed a clean UI displaying three key data points: **Room Name**, **Course Name**, and the **List of Enrolled Students**.
- **Data Rendering:** Used Thymeleaf `th:each` to dynamically render real-time data from the database onto the browser at the `/classrooms` endpoint.

Conclusion

- **Current Status:** The Classroom View feature is 100% functional across both Backend and UI layers, successfully fetching and displaying data from MySQL.
- **GitHub Workflow:** All updates have been committed and pushed to the `pheakdey` branch. Pull Request #35 has been reopened for final review and merging.